Ahmad Nouh

AI / ML Engineer | LLM Applications, Agentic AI & Search

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PROFESSIONAL SUMMARY

AI/ML Engineer with six years of experience in NLP, LLM applications, search, and Python backend systems. Cut a core AI pipeline from about 15 minutes to under 3 and built a spam-classification service handling up to 30,000 requests per day. Delivered event recommendations and contributed to agentic data enrichment, with experience in model training, evaluation, and deployment.

TECHNICAL SKILLS

ML and NLP: TensorFlow/Keras, PyTorch, scikit-learn, Hugging Face Transformers, PEFT/QLoRA, Arabic NLP, text classification, NER, OCR, pandas, NumPy.

LLMs and search: LangGraph, LangChain, retrieval-augmented generation (RAG), tool calling, MCP, Azure OpenAI, Gemini, embeddings, semantic and hybrid search, PostgreSQL/pgvector, Neo4j.

Backend: Python, SQL, FastAPI, Django, Flask, REST APIs, Celery, RabbitMQ, Redis; Go, Java, Node.js, TypeScript.

MLOps and cloud: MLflow, DVC, model evaluation, Docker, Git, CI/CD, Azure Machine Learning, AWS EC2 GPUs, Grafana.

PROFESSIONAL EXPERIENCE

Blink

Nov 2025 - Jul 2026

AI Engineer

Remote | Company HQ: Falls Church, Virginia, US

- Built session recommendations and attendee matchmaking for an event platform using content-based filtering. Developed Python/FastAPI services with PostgreSQL/pgvector, RabbitMQ, automated tests, and staged releases.
- Designed and evaluated a beta, feature-flagged flight-creation agent with LangGraph. Connected five tools for booking, attendee search, location lookup, and document parsing within an event platform.
- Built a document-extraction workflow with Docling and Gemini 2.5 Flash to turn flight-ticket and hotel PDFs and images into structured data.
- Evaluated self-hosting Gemma 3 12B on AWS EC2 G5 GPUs with Ollama. Benchmarked time to first token, response latency, throughput, memory use, and GPU utilization.

MENT

Oct 2023 - Oct 2025

AI Engineer

Remote | Company HQ: Falls Church, Virginia, US

- Cut AI pipeline runtime from **about 15 minutes to under 3 minutes** by parallelizing asynchronous LLM extraction calls and moving schema mapping and validation into Python.
- Improved matching of collected data to the correct professional profiles by approximately 33% versus the prior system, addressing people with identical or similar names.
- Contributed to multi-agent workflows that enrich professional profiles in n8n. Integrated MCP tools, RabbitMQ retries, and Grafana tracing.
- Built a Neo4j relationship graph to generate contact insights, including shared interests, employers, and connections through mutual contacts.
- Built Django APIs for generating insights and message drafts, with contact import/export. Developed Python/Celery enrichment services and integrated Azure OpenAI.

PROFESSIONAL EXPERIENCE CONTINUED

Lableb

Aug 2020 - Sep 2023

AI Engineer

Remote | Company HQ: Dubai, UAE

- Built a production B2B spam-classification service handling **up to 30,000 requests per day** and filtering **about 90% of spam queries**.
- Compared n-gram and neural models for Arabic spelling correction. Adopted KenLM for faster processing with comparable results.
- Trained and evaluated Arabic named entity recognition (NER), part-of-speech (POS) tagging, transliteration, and spelling-correction models. Used TensorFlow/Keras and Azure Machine Learning; tracked model metrics and versioned data and artifacts with DVC.
- Built NLP microservices in Python, Go, and Java; researched Arabic semantic search with Hugging Face models.

AI Engineer Intern | Lableb | Apr 2020 - Jul 2020

Code Experts

Oct 2018 - Jan 2020

Web Developer

Damascus, Syria | On-site

- Built Angular and Ionic interfaces for video-on-demand and taxi-booking products; contributed Node.js backend work and a WebSocket chat feature connecting drivers and riders.

SELECTED PROJECTS

Second Memory | Hybrid search and knowledge assistant

Built a self-hosted knowledge assistant with FastAPI, LangGraph, PostgreSQL/pgvector, and Flutter. Combined vector and full-text search with Reciprocal Rank Fusion (RRF); added account-scoped retrieval, streamed answers, validated references, CI tests, and Recall@k/MRR evaluation.

SILA-4B | Arabic LLM fine-tuning and evaluation

Built a Qwen3-4B Arabic tool-calling experiment using PyTorch, Transformers, and PEFT/QLoRA on an 8 GB GPU. Prepared 6,600 synthetic examples with train/validation overlap checks. Built an evaluation workflow comparing base and fine-tuned models on 100 Arabic and 500 English BFCL tool-calling cases.

End-to-End Topic Classification

Built an MLflow pipeline for training, tuning, prediction, and model evaluation. Used Snorkel to inspect performance on emoji and short tweets. Added Streamlit views to explore data, evaluate models, and run predictions.

EDUCATION

Bachelor's degree in Information Technology Engineering
Damascus University | 2015 - 2021 | Artificial Intelligence specialization

LANGUAGES

Arabic: Native or bilingual | English: Professional working | Turkish: Elementary

SELECTED CERTIFICATIONS

DeepLearning.AI TensorFlow Developer | July 2021
Introduction to Machine Learning in Production | DeepLearning.AI